IF A SCHEME IS ETALE OVER THE BASE AND ITS TARGET IS UNRAMIFIED OVER THE BASE, THE MORPHISM BETWEEN THEM IS ETALE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. The statement span contains an author 'reference' block reading 'See for example \cite[Remark 18.30 (2).]{GWII}.' – a pointer to the literature, not part of the assertion. The counted proof is self-contained and does not use the citation; the card is assembled as-is and no bibliography entry is invented for it.

Allowed standard prerequisites (editorial).

- Etaleness is stable under composition. Lemma \ref{lemma-composition-etale}, cited at proof lines 8471-8472. Combines the two etale arrows into the conclusion.
- Etaleness is stable under base change. Lemma \ref{lemma-base-change-etale}, cited at proof lines 8473-8474. Shows the projection of the fibre product is etale.
- The diagonal of an unramified morphism is an open immersion. Lemma \ref{lemma-diagonal-unramified-morphism}, cited at proof lines 8483-8484. Identifies the first arrow as a base change of an open immersion.
- Open immersions are etale, and are stable under base change. Lemma \ref{lemma-open-immersion-etale} and Schemes, Lemma \ref{schemes-lemma-base-change-immersion}, cited at proof lines 8485-8488. Gives etaleness of the first arrow.

Source reference. Original lemma 0HB1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21581–21581. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02GN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3234–3234. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02GO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3235–3235. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02GE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3225–3225. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01JY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2031–2031. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02GP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3236–3236. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original reference cite-GWII: sources/stacks-morphisms/pinned-original-source.tex, source lines 8458–8458. Bibliographic pointer appearing in the counted text; it carries no mathematical argument.

Lemma 0.2. *Let $f : X \to Y$ be a morphism of schemes over S. If X is étale over S and Y is unramified over S, then f is étale.*

Proof. Consider the factorization $X \to X \times_S Y \to Y$, where the first arrow is given by id_X and f and the second arrow is the projection. We claim both arrows are étale and hence f is étale by Lemma 02GN. Namely, the projection is étale as it is the base change of $X \to S$, see Lemma 02GO. The first arrow is the base change of the diagonal morphism $Y \to Y \times_S Y$ because the square

$$\begin{array}{ccc} X & \longrightarrow & X \times_S Y \\ \downarrow & & \downarrow \\ Y & \longrightarrow & Y \times_S Y \end{array}$$

is cartesian. The diagonal $Y \to Y \times_S Y$ is an open immersion by Lemma 02GE. The base change of an open immersion is an open immersion (Schemes, Lemma 01JY) and an open immersion is étale (Lemma 02GP). □